

Krishnaa Nair

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EDUCATION

- Vellore Institute of Technology, Chennai, India** Aug 2023 - Present (2027)
B.Tech in Computer Science and Engineering - AI & Robotics | CGPA: 8.35 / 10
Relevant Coursework: Machine Learning, Deep Learning, Computer Vision, DSA, Embedded Systems, Database Systems, Statistics
- The Indian High School, Dubai, UAE** Apr 2021 - Mar 2023
CBSE Class XII: 89.2% | Artificial Intelligence: 100%

SKILLS

Languages: Python, R, C, C++, Java, JavaScript, HTML, CSS, MATLAB, SQL

AI, ML & Data: PyTorch, TensorFlow, Keras, Scikit-learn, OpenCV, YOLOv8, MONAI, Pandas, NumPy, Matplotlib

Frameworks & Platforms: React, FastAPI, Ollama, Vosk, ElevenLabs, Raspberry Pi, Arduino, Git, GitHub, VS Code

Domains: Machine Learning, Deep Learning, Computer Vision, Generative AI, Voice Interfaces, Human-Robot Interaction, IoT, Embedded AI, Data Analysis

PROJECTS

• DeskPilot AI - Agentic Desktop Automation Platform:

Architected an end-to-end AI automation system that converts spoken or typed objectives into executable, multi-application desktop workflows. Integrated a React command centre, FastAPI orchestration layer, local Ollama reasoning, Vosk speech recognition and ElevenLabs feedback into a 6-technology stack; benchmarked an 8-second end-to-end execution cycle and built failure logging for workflow reliability. Selected among 40 of 5,000 applicants (top 0.8%).

• Smart Mirror - Patent-Filed Edge AI Health & Automation System:

Designed and developed a patent-filed smart mirror that unifies real-time skincare analysis, biometric monitoring, environmental sensing and voice-controlled home automation on Raspberry Pi 5. Engineered 4 integrated subsystems: YOLOv8 acne analysis, MAX30102 heart-rate/SpO2 monitoring, DHT11 climate sensing and relay-based appliance control, supported by camera, microphone and speaker interfaces. Indian Patent Application No. 202641022498.

• Smart Robot Dog Companion - Multimodal Social Robotics Platform:

Engineered a Raspberry Pi 5-based social robot that interprets touch, voice, movement and orientation to reproduce responsive companion behaviour. Integrated a 5-inch display, dual servo ears, dual BO motors, L298N motor control, IMU, touch sensor, microphone and camera; programmed real-time affection responses, command-driven locomotion, expressive ear movement and flip-distress safety behaviour.

• Multimodal MRI Brain Tumour Segmentation - 3D Medical AI Pipeline:

Built an end-to-end BraTS 2021 medical-imaging pipeline for voxel-level brain tumour segmentation using MONAI U-Net and SegResNet architectures. Standardised 3 MRI sequences into 128 x 128 x 128 volumes and implemented preprocessing, augmentation, model training and inference workflows for reproducible 3D tumour-region analysis.

• Voice-Controlled 4-DOF Robotic Arm - Embedded Manipulation System:

Designed and built a 4-degree-of-freedom robotic manipulator that translates spoken commands into coordinated multi-joint motion. Programmed Arduino Uno control across 4 servo-driven axes, integrated HC-05 Bluetooth command transmission, calibrated joint limits and motion sequences, and implemented repeatable voice-triggered positioning routines for hands-free robotic control.

LEADERSHIP

- General Secretary - Event Managers Club, VIT Chennai** Nov 2025 - Present
Direct annual event calendars, budgeting, resource allocation and cross-team coordination across operations, design, finance and stakeholder teams.
- Overall Head - Vibrance 2026 Events Fest, VIT Chennai** 2026
Owned end-to-end scheduling, venue readiness, team deployment, approvals and live issue escalation across concurrent event verticals.
- Overall Coordinator - TechnoVIT 2025 Events Fest, VIT Chennai** 2025
Coordinated logistics, evaluation workflows and inter-team dependencies across concurrent technical events and delivery timelines.
- Head of AI/ML - CloudOps, VIT Chennai** Sep 2025 - Present
Lead AI/ML technical sessions, hackathon activities, peer mentoring and practical cloud-native machine-learning prototypes.
- School Football Team Member - The Indian High School, Dubai**
Event-fest execution experience includes coordination across 600+ events serving 3,000+ attendees.

ACHIEVEMENTS

- 2nd Prize - Industry Academia Innovators Conclave (IAIC 2025):**
Recognised for an industry-focused innovation integrating AI, sensing and automation.
- Selected Delegate - Harvard Project for Asian and International Relations (HPAIR):**
Selected to participate in an international student conference focused on leadership and global affairs.
- Perfect Score - CBSE Class XII Artificial Intelligence (100%):**
Received CBSE UAE Sahodaya recognition for academic excellence in Artificial Intelligence.

CERTIFICATIONS

- Generative AI - IBM**
- Machine Learning with Python - Indian Institute of Technology Kanpur**
- Management Program - Indian Institute of Management Udaipur**